

Calculer les limites suivantes:

$$\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7) ; \lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3 ; \lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5} ; \lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3}$$

$$\lim_{x \rightarrow +\infty} \frac{x^2+5x}{(2x+3)^2} ; \lim_{x \rightarrow +\infty} \frac{2x^2-9x+5}{2x^2-(2x+1)^2} ; \lim_{x \rightarrow +\infty} \frac{(1-\sqrt{2})x^2+3x-5}{x^2+x+2} ; \lim_{x \rightarrow -\infty} \left( \frac{\sqrt{2}x^2}{x+1} - \sqrt{3}x \right)$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8} ; \lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2-5x}{8x^2+7}} ; \lim_{x \rightarrow +\infty} \sqrt{\frac{1}{2x-5}} ; \lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3-1}{2x^3+7}}$$

$$\lim_{x \rightarrow \sqrt{2}} \frac{x^2-2}{x-\sqrt{2}} ; \lim_{x \rightarrow 3} \frac{x^2-5x+6}{9-x^2} ; \lim_{x \rightarrow -5} \frac{x^2+6x+5}{x^2+4x-5} ; \lim_{x \rightarrow 2} \frac{x^3-8}{x^2-2x} ; \lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x-x^2}$$

$$\lim_{x \rightarrow 4} \frac{\sqrt{2x+1}-3}{\sqrt{x}-2} ; \lim_{x \rightarrow -1} \frac{2x^3+3x^2+4x+3}{x^2+3x+2} ; \lim_{x \rightarrow 3} \frac{x^2-x-6}{\sqrt{x+6}-3} ; \lim_{x \rightarrow 3} \frac{\sqrt{2x+3}-\sqrt{x+6}}{\sqrt{x-1}-\sqrt{8-2x}}$$

$$\lim_{x \rightarrow 4} \frac{\sqrt{x+5}-\sqrt{x}-1}{\sqrt{x+12}-\sqrt{x}-2} ; \lim_{x \rightarrow 3} \frac{7-5x}{(x-3)^2} ; \lim_{x \rightarrow -1^-} \frac{x+1}{\sqrt{x^2-1}} ; \lim_{x \rightarrow 1^-} \frac{x^2-x-3}{x^2+2x-3}$$

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^2-3x+2}}{x-1} ; \lim_{x \rightarrow 1^-} \frac{8x-1}{\sqrt{1-x}} ; \lim_{x \rightarrow 4^+} \frac{x}{2-\sqrt{x}} ; \lim_{x \rightarrow 4^+} \frac{\sqrt{x+12}-\sqrt{3x-4}}{x^2-8x+16}$$

$$\lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2-4}}$$

$$\lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x-2} ; \lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)} ; \lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}}$$

$$\lim_{x \rightarrow -3^-} \frac{x^3}{|x^2 - 9|} ; \lim_{x \rightarrow (\frac{1}{2})^+} \frac{3x^2 - x - 1}{4x^2 - 1} ; \lim_{x \rightarrow +\infty} (2x - 5\sqrt{x}) ; \lim_{x \rightarrow -\infty} (\sqrt{9x^2 + 3x - 5} + x + 3)$$

$$\lim_{x \rightarrow +\infty} (\sqrt{x+3} - x) ; \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4) ; \lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x})$$

$$\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3) ; \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x} ; \lim_{x \rightarrow +\infty} \sqrt{\frac{4x^2 + 3}{x^2 + 8x}}$$

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2} ; \lim_{x \rightarrow +\infty} \frac{x + 6\sqrt{x}}{2x + \sqrt{x} - 3} ; \lim_{x \rightarrow +\infty} \frac{1}{x} (\sqrt{3x^2 + 1} - \sqrt{x^2 + x}) ; \lim_{x \rightarrow +\infty} (\sqrt{x^4 - x^3} - x^2)$$

$$\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2 + x} - 3x^2 + x}{5\sqrt{x^2 + 3}} ; \lim_{x \rightarrow +\infty} \left( \sqrt{\frac{x^3}{x+1}} - x \right) ; \lim_{x \rightarrow +\infty} \left( \frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right)$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7)$$

On a :

$$\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7) = \lim_{x \rightarrow -\infty} 4x^5$$

Donc :

$$\lim_{x \rightarrow -\infty} (4x^5 - 3x^2 + 7) = -\infty$$

Car :

$$\lim_{x \rightarrow -\infty} x^5 = -\infty \quad \text{et} \quad 4 > 0$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3 &= \lim_{x \rightarrow +\infty} (-2x^4)^3 \\ &= \lim_{x \rightarrow +\infty} -8x^{12} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} (-2x^4 + x^3 - 3x - 1)^3 = -\infty$$

$$\text{Car : } \lim_{x \rightarrow -\infty} x^{12} = +\infty \quad \text{et} \quad -8 < 0$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5} &= \lim_{x \rightarrow -\infty} \frac{x}{x^2} \\ &= \lim_{x \rightarrow -\infty} \frac{1}{x} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -\infty} \frac{x+3}{x^2-5} = 0$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3} &= \lim_{x \rightarrow +\infty} \frac{-x^3}{x^3} \\ &= \lim_{x \rightarrow +\infty} -1 \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{(1-x)^3}{1+x+x^3} = -1$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 5x}{(2x + 3)^2}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{x^2 + 5x}{(2x + 3)^2} &= \lim_{x \rightarrow +\infty} \frac{x^2}{4x^2} \\ &= \lim_{x \rightarrow +\infty} \frac{1}{4} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 5x}{(2x + 3)^2} = \frac{1}{4}$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - 9x + 5}{2x^2 - (2x + 1)^2}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{2x^2 - 9x + 5}{2x^2 - (2x + 1)^2} &= \lim_{x \rightarrow +\infty} \frac{2x^2 - 9x + 5}{2x^2 - (2x + 1)^2} \\ &= \lim_{x \rightarrow +\infty} \frac{2x^2 - 9x + 5}{-2x^2 - 4x - 1} \end{aligned}$$

Donc :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{2x^2 - 9x + 5}{2x^2 - (2x + 1)^2} &= \lim_{x \rightarrow +\infty} \frac{2x^2}{-2x^2} \\ &= \lim_{x \rightarrow +\infty} -1 \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - 9x + 5}{2x^2 - (2x + 1)^2} = -1$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{(1 - \sqrt{2})x^2 + 3x - 5}{x^2 + x + 2}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{(1 - \sqrt{2})x^2 + 3x - 5}{x^2 + x + 2} &= \lim_{x \rightarrow +\infty} \frac{(1 - \sqrt{2})x^2}{x^2} \\ &= \lim_{x \rightarrow +\infty} (1 - \sqrt{2}) \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{(1 - \sqrt{2})x^2 + 3x - 5}{x^2 + x + 2} = (1 - \sqrt{2})$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} \left( \frac{\sqrt{2} x^2}{x+1} - \sqrt{3}x \right)$$

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} \left( \frac{\sqrt{2} x^2}{x+1} - \sqrt{3}x \right) &= \lim_{x \rightarrow -\infty} \frac{\sqrt{2}x^2 - \sqrt{3}x(x+1)}{x+1} \\ &= \lim_{x \rightarrow -\infty} \frac{(\sqrt{2} - \sqrt{3})x^2 - \sqrt{3}x}{x+1} \\ &= \lim_{x \rightarrow -\infty} \frac{(\sqrt{2} - \sqrt{3})x^2}{x} \\ &= \lim_{x \rightarrow -\infty} (\sqrt{2} - \sqrt{3})x \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -\infty} \left( \frac{\sqrt{2} x^2}{x+1} - \sqrt{3}x \right) = +\infty$$

Car :

$$\lim_{x \rightarrow -\infty} x = -\infty \quad \text{et} \quad (\sqrt{2} - \sqrt{3}) < 0$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8}$$

On a :

$$\lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8} = \lim_{x \rightarrow +\infty} \sqrt{x^3 \left( 1 - \frac{2}{x} + \frac{3}{x^2} - \frac{8}{x^3} \right)}$$

Donc :

$$\lim_{x \rightarrow +\infty} \sqrt{x^3 - 2x^2 + 3x - 8} = +\infty$$

Car :

$$\left\{ \begin{array}{l} \lim_{x \rightarrow +\infty} \frac{2}{x} = \lim_{x \rightarrow +\infty} \frac{3}{x^2} = \lim_{x \rightarrow +\infty} \frac{8}{x^3} = 0 \\ \lim_{x \rightarrow +\infty} x^3 = +\infty \\ \lim_{X \rightarrow +\infty} \sqrt{X} = +\infty \end{array} \right.$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2 - 5x}{8x^2 + 7}}$$

On a :

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2 - 5x}{8x^2 + 7}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{x^2 \left(2 - \frac{5}{x}\right)}{x^2 \left(8 + \frac{7}{x^2}\right)}}$$

$$= \lim_{x \rightarrow -\infty} \sqrt{\frac{\left(2 - \frac{5}{x}\right)}{\left(8 + \frac{7}{x^2}\right)}}$$

Donc :

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{2x^2 - 5x}{8x^2 + 7}} = \frac{1}{2}$$

Car :

$$\lim_{x \rightarrow -\infty} \frac{5}{x} = \lim_{x \rightarrow -\infty} \frac{7}{x^2} = 0$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{1}{2x - 5}}$$

On a :

$$\lim_{x \rightarrow +\infty} \frac{1}{2x - 5} = 0^+$$

et

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0$$

Donc :

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{1}{2x - 5}} = 0$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3 - 1}{2x^3 + 7}}$$

On a :

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3 - 1}{2x^3 + 7}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{x^3 \left(5 - \frac{1}{x^3}\right)}{x^3 \left(2 + \frac{7}{x^3}\right)}}$$

Donc :

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3 - 1}{2x^3 + 7}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{\left(5 - \frac{1}{x^3}\right)}{\left(2 + \frac{7}{x^3}\right)}}$$

Donc :

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{5x^3 - 1}{2x^3 + 7}} = \frac{\sqrt{10}}{2}$$

Car :

$$\lim_{x \rightarrow -\infty} \frac{1}{x^3} = \lim_{x \rightarrow -\infty} \frac{7}{x^3} = 0$$

➤ Calculons :

$$\lim_{x \rightarrow \sqrt{2}} \frac{x^2 - 2}{x - \sqrt{2}}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow \sqrt{2}} \frac{x^2 - 2}{x - \sqrt{2}} &= \lim_{x \rightarrow \sqrt{2}} \frac{x^2 - (\sqrt{2})^2}{x - \sqrt{2}} \\ &= \lim_{x \rightarrow \sqrt{2}} \frac{(x - \sqrt{2})(x + \sqrt{2})}{x - \sqrt{2}} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow \sqrt{2}} \frac{x^2 - 2}{x - \sqrt{2}} = \lim_{x \rightarrow \sqrt{2}} (x + \sqrt{2})$$

Donc :

$$\lim_{x \rightarrow \sqrt{2}} \frac{x^2 - 2}{x - \sqrt{2}} = 2\sqrt{2}$$

➤ Calculons :

$$\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{9 - x^2}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{9 - x^2} &= \lim_{x \rightarrow 3} \frac{(x - 2)(x - 3)}{(3 - x)(3 + x)} \\ &= \lim_{x \rightarrow 3} \frac{-(x - 2)}{(3 + x)} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{9 - x^2} = -\frac{1}{6}$$

➤ Calculons :

$$\lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5}$$

On a :

$$\begin{aligned}\lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5} &= \lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5} \\ &= \lim_{x \rightarrow -5} \frac{(x + 1)(x + 5)}{(x - 1)(x + 5)} \\ &= \lim_{x \rightarrow -5} \frac{(x + 1)}{(x - 1)}\end{aligned}$$

Donc :

$$\lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x^2 + 4x - 5} = \frac{2}{3}$$

➤ Calculons :

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x}$$

On a :

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} = \lim_{x \rightarrow 2} \frac{x^3 - 2^3}{x(x - 2)}$$

Donc :

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{x(x - 2)} \\ &= \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x}\end{aligned}$$

Donc :

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} = 6$$

➤ Calculons :

$$\lim_{x \rightarrow 0} \frac{\sqrt{x + 4} - 2}{x - x^2}$$

On a :

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sqrt{x + 4} - 2}{x - x^2} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x + 4} - 2)(\sqrt{x + 4} + 2)}{x(1 - x)(\sqrt{x + 4} + 2)} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(1 - x)(\sqrt{x + 4} + 2)} \\ &= \lim_{x \rightarrow 0} \frac{1}{(1 - x)(\sqrt{x + 4} + 2)}\end{aligned}$$

Donc :

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x - x^2} = \frac{1}{4}$$

➤ Calculons :

$$\lim_{x \rightarrow 4} \frac{\sqrt{2x+1} - 3}{\sqrt{x} - 2}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{\sqrt{2x+1} - 3}{\sqrt{x} - 2} &= \lim_{x \rightarrow 4} \frac{(\sqrt{2x+1} - 3)(\sqrt{2x+1} + 3)(\sqrt{x} + 2)}{(\sqrt{x} - 2)(\sqrt{x} + 2)(\sqrt{2x+1} + 3)} \\ &= \lim_{x \rightarrow 4} \frac{(2x + 1 - 9)(\sqrt{x} + 2)}{(x - 4)(\sqrt{2x+1} + 3)} \\ &= \lim_{x \rightarrow 4} \frac{2(x - 4)(\sqrt{x} + 2)}{(x - 4)(\sqrt{2x+1} + 3)} \\ &= \lim_{x \rightarrow 4} \frac{2(\sqrt{x} + 2)}{\sqrt{2x+1} + 3} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 4} \frac{\sqrt{2x+1} - 3}{\sqrt{x} - 2} = \frac{4}{3}$$

➤ Calculons :

$$\lim_{x \rightarrow -1} \frac{2x^3 + 3x^2 + 4x + 3}{x^2 + 3x + 2}$$

On a :

$$\begin{array}{r|l}
 2x^3 + 3x^2 + 4x + 3 & x + 1 \\
 - 2x^3 + 2x^2 & \hline
 \hline
 & x^2 + 4x \\
 - & x^2 + x \\
 \hline
 & 3x + 3 \\
 - & 3x + 3 \\
 \hline
 & 0
 \end{array}$$

Donc :

$$\begin{aligned}
 \lim_{x \rightarrow -1} \frac{2x^3 + 3x^2 + 4x + 3}{x^2 + 3x + 2} &= \lim_{x \rightarrow -1} \frac{(2x^2 + x + 3)(x + 1)}{(x + 1)(x + 2)} \\
 &= \lim_{x \rightarrow -1} \frac{2x^2 + x + 3}{x + 2}
 \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -1} \frac{2x^3 + 3x^2 + 4x + 3}{x^2 + 3x + 2} = 4$$

➤ Calculons :

$$\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{\sqrt{x + 6} - 3}$$

On a :

$$\begin{aligned}
 \lim_{x \rightarrow 3} \frac{x^2 - x - 6}{\sqrt{x + 6} - 3} &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 2)(\sqrt{x + 6} + 3)}{(\sqrt{x + 6} - 3)(\sqrt{x + 6} + 3)} \\
 &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 2)(\sqrt{x + 6} + 3)}{x - 3} \\
 &= \lim_{x \rightarrow 3} (x + 2)(\sqrt{x + 6} + 3)
 \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{\sqrt{x + 6} - 3} = 30$$

➤ Calculons :

$$\lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - \sqrt{x+6}}{\sqrt{x-1} - \sqrt{8-2x}}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - \sqrt{x+6}}{\sqrt{x-1} - \sqrt{8-2x}} &= \lim_{x \rightarrow 3} \frac{(\sqrt{2x+3} - \sqrt{x+6})(\sqrt{2x+3} + \sqrt{x+6})(\sqrt{x-1} + \sqrt{8-2x})}{(\sqrt{x-1} - \sqrt{8-2x})(\sqrt{x-1} + \sqrt{8-2x})(\sqrt{2x+3} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{(x-3)(\sqrt{x-1} + \sqrt{8-2x})}{3(x-3)(\sqrt{2x+3} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{\sqrt{x-1} + \sqrt{8-2x}}{3(\sqrt{2x+3} + \sqrt{x+6})} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - \sqrt{x+6}}{\sqrt{x-1} - \sqrt{8-2x}} = \frac{\sqrt{2}}{9}$$

➤ Calculons :

$$\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2} &= \lim_{x \rightarrow 4} \frac{\sqrt{x+5} - (\sqrt{x} + 1)}{\sqrt{x+12} - (\sqrt{x} + 2)} \\ &= \lim_{x \rightarrow 4} \frac{[\sqrt{x+5} - (\sqrt{x} + 1)][\sqrt{x+5} + (\sqrt{x} + 1)][\sqrt{x+12} + (\sqrt{x} + 2)]}{[\sqrt{x+12} - (\sqrt{x} + 2)][\sqrt{x+12} + (\sqrt{x} + 2)][\sqrt{x+5} + (\sqrt{x} + 1)]} \\ &= \lim_{x \rightarrow 4} \frac{-2[\sqrt{x} - 2][\sqrt{x+12} + (\sqrt{x} + 2)]}{-4[\sqrt{x} - 2][\sqrt{x+5} + (\sqrt{x} + 1)]} \\ &= \lim_{x \rightarrow 4} \frac{\sqrt{x+12} + (\sqrt{x} + 2)}{2[\sqrt{x+5} + (\sqrt{x} + 1)]} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - \sqrt{x} - 1}{\sqrt{x+12} - \sqrt{x} - 2} = \frac{2}{3}$$

➤ Calculons :

$$\lim_{x \rightarrow 3} \frac{7 - 5x}{(x - 3)^2}$$

On a :

$$\lim_{x \rightarrow 3} 7 - 5x = -8$$

et

$$\lim_{x \rightarrow 3} (x - 3)^2 = 0^+$$

Donc :

$$\lim_{x \rightarrow 3} \frac{7 - 5x}{(x - 3)^2} = -\infty$$

Car :

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty \quad \text{et} \quad -8 < 0$$

➤ Calculons :

$$\lim_{x \rightarrow -1^-} \frac{x + 1}{\sqrt{x^2 - 1}}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow -1^-} \frac{x + 1}{\sqrt{x^2 - 1}} &= \lim_{x \rightarrow -1^-} \frac{-\sqrt{(x + 1)^2}}{\sqrt{(x - 1)(x + 1)}} \\ &= \lim_{x \rightarrow -1^-} -\sqrt{\frac{x + 1}{x - 1}} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -1^-} \frac{x + 1}{\sqrt{x^2 - 1}} = 0$$

Car :

$$\begin{cases} \lim_{x \rightarrow -1^-} x + 1 = 0^- \\ \lim_{x \rightarrow -1^-} x - 1 = -2 \\ \lim_{x \rightarrow 0^+} \sqrt{x} = 0 \end{cases}$$

➤ Calculons :

$$\lim_{x \rightarrow 1^-} \frac{x^2 - x - 3}{x^2 + 2x - 3}$$

Le discriminant du trinôme  $x^2 + 2x - 3$  est :  $\Delta = 2^2 - 4 \times 1 \times (-3) = 16$ . Donc le trinôme  $x^2 + 2x - 3$  admet deux racines distinctes qui sont :  $x_1 = -3$  et  $x_2 = 1$ .

Le tableau de signe du trinôme  $x^2 + 2x - 3$  est :

| $x$            | $-\infty$ | $-3$ | $1$ | $+\infty$ |     |
|----------------|-----------|------|-----|-----------|-----|
| $x^2 + 2x - 3$ | $+$       | $0$  | $-$ | $0$       | $+$ |

D'où :

$$\lim_{x \rightarrow 1^-} x^2 + 2x - 3 = 0^-$$

On a :

$$\lim_{x \rightarrow 1^-} x^2 - x - 3 = -3$$

Donc :

$$\lim_{x \rightarrow 1^-} \frac{x^2 - x - 3}{x^2 + 2x - 3} = +\infty$$

$$\text{Car } \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \quad \text{et} \quad -3 < 0$$

➤ Calculons :

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^2 - 3x + 2}}{x - 1}$$

On a :

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^2 - 3x + 2}}{x - 1} = \lim_{x \rightarrow 1^-} \frac{\sqrt{(x - 1)(x - 2)}}{-\sqrt{(x - 1)^2}}$$

$$\text{Donc :} \quad = \lim_{x \rightarrow 1^-} -\sqrt{\frac{x - 2}{x - 1}}$$

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^2 - 3x + 2}}{x - 1} = -\infty$$

Car :

$$\begin{cases} \lim_{x \rightarrow 1^-} x - 2 = -1 \\ \lim_{x \rightarrow 1^-} x - 1 = 0^- \\ \lim_{x \rightarrow 0^-} \frac{-1}{X} = +\infty \end{cases}$$

➤ Calculons :

$$\lim_{x \rightarrow 1^-} \frac{8x - 1}{\sqrt{1 - x}}$$

On a :

$$\lim_{x \rightarrow 1^-} 8x - 1 = 7$$

et

$$\lim_{x \rightarrow 1^-} \sqrt{1 - x} = 0^+$$

Donc :

$$\lim_{x \rightarrow 1^-} \frac{8x - 1}{\sqrt{1 - x}} = +\infty$$

Car :

$$\lim_{x \rightarrow 0^+} \frac{7}{X} = +\infty$$

➤ Calculons :

$$\lim_{x \rightarrow 4^+} \frac{x}{2 - \sqrt{x}}$$

On a :

$$\lim_{x \rightarrow 4^+} x = 4$$

et

$$\lim_{x \rightarrow 4^+} 2 - \sqrt{x} = 0^-$$

Donc :

$$\lim_{x \rightarrow 4^+} \frac{x}{2 - \sqrt{x}} = -\infty$$

Car :

$$\lim_{x \rightarrow 0^-} \frac{4}{X} = -\infty$$

➤ Calculons :

$$\lim_{x \rightarrow 4^+} \frac{\sqrt{x+12} - \sqrt{3x-4}}{x^2 - 8x + 16}$$

On a :

$$\lim_{x \rightarrow 4^+} \sqrt{x+12} - \sqrt{3x-4} = 4 - 2\sqrt{2}$$

et

$$\begin{aligned} \lim_{x \rightarrow 4^+} x^2 - 8x + 16 &= \lim_{x \rightarrow 4^+} (x-4)^2 \\ &= 0^+ \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 4^+} \frac{\sqrt{x+12} - \sqrt{3x-4}}{x^2 - 8x + 16} = +\infty$$

Car

$$\lim_{x \rightarrow 0^+} \frac{4 - 2\sqrt{2}}{X} = +\infty$$

➤ Calculons :

$$\lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2 - 4}}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2 - 4}} &= \lim_{x \rightarrow 2^+} \frac{\sqrt{x} - \sqrt{2}}{\sqrt{x^2 - 4}} + \frac{\sqrt{x-2}}{\sqrt{x^2 - 4}} \\ &= \lim_{x \rightarrow 2^+} \frac{(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})}{\sqrt{x^2 - 4}(\sqrt{x} + \sqrt{2})} + \frac{\sqrt{x-2}}{\sqrt{(x-2)(x+2)}} \\ &= \lim_{x \rightarrow 2^+} \frac{\sqrt{x-2}}{\sqrt{(x+2)}(\sqrt{x} + \sqrt{2})} + \frac{1}{\sqrt{(x+2)}} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 2^+} \frac{\sqrt{x} + \sqrt{x-2} - \sqrt{2}}{\sqrt{x^2 - 4}} = \frac{1}{2}$$

➤ Calculons :

$$\lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x-2}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x-2} &= \lim_{x \rightarrow 2^-} \left( \frac{\sqrt{4-x^2}}{x-2} + 1 \right) \\ &= \lim_{x \rightarrow 2^-} \left( \frac{\sqrt{(2-x)(2+x)}}{-\sqrt{(x-2)^2}} + 1 \right) \\ &= \lim_{x \rightarrow 2^-} \left( -\sqrt{\frac{2+x}{2-x}} + 1 \right) \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 2^-} \frac{\sqrt{4-x^2} + x - 2}{x-2} = -\infty$$

Car :

$$\begin{cases} \lim_{x \rightarrow 2^-} 2-x = 0^+ \\ \lim_{X \rightarrow 0^+} \frac{4}{X} = +\infty \\ \lim_{Y \rightarrow +\infty} \sqrt{Y} = +\infty \end{cases}$$

➤ Calculons :

$$\lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)} &= \lim_{x \rightarrow -1} \frac{x^3 - (-1)^3}{(x+1)^3(x-2)} \\ &= \lim_{x \rightarrow -1} \frac{(x+1)(x^2 - x + 1)}{(x+1)^3(x-2)} \\ &= \lim_{x \rightarrow -1} \frac{x^2 - x + 1}{(x+1)^2(x-2)} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -1} \frac{x^3 + 1}{(x+1)^3(x-2)} = -\infty$$

Car :

$$\begin{cases} \lim_{x \rightarrow -1} x^2 - x + 1 = 3 \\ \lim_{x \rightarrow -1} (x+1)^2(x-2) = 0^- \\ \lim_{X \rightarrow 0^-} \frac{1}{X} = -\infty \end{cases}$$

➤ Calculons :

$$\lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}} &= \lim_{x \rightarrow 0^+} \frac{(2 - \sqrt{x^2 + 4})(2 + \sqrt{x^2 + 4})}{\sqrt{x}(1 - \sqrt{2x})(2 + \sqrt{x^2 + 4})} \\ &= \lim_{x \rightarrow 0^+} \frac{-x^2 \times \sqrt{x}}{\sqrt{x} \times \sqrt{x}(1 - \sqrt{2x})(2 + \sqrt{x^2 + 4})} \\ &= \lim_{x \rightarrow 0^+} \frac{-x\sqrt{x}}{(1 - \sqrt{2x})(2 + \sqrt{x^2 + 4})} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow 0^+} \frac{2 - \sqrt{x^2 + 4}}{\sqrt{x} - \sqrt{2x^2}} = 0$$

Car :

$$\begin{cases} \lim_{x \rightarrow 0^+} -x\sqrt{x} = 0 \\ \lim_{x \rightarrow 0^+} (1 - \sqrt{2x})(2 + \sqrt{x^2 + 4}) = 4 \end{cases}$$

➤ Calculons :

$$\lim_{x \rightarrow -3^-} \frac{x^3}{|x^2 - 9|}$$

On a :

$$\lim_{x \rightarrow -3^-} x^3 = -27$$

et

$$\lim_{x \rightarrow -3^-} |x^2 - 9| = 0^+$$

Donc :

$$\lim_{x \rightarrow -3^-} \frac{x^3}{|x^2 - 9|} = -\infty$$

Car :

$$\lim_{x \rightarrow 0^+} \frac{-27}{X} = -\infty$$

➤ Calculons :

$$\lim_{x \rightarrow (\frac{1}{2})^+} \frac{3x^2 - x - 1}{4x^2 - 1}$$

On a :

$$\lim_{x \rightarrow (\frac{1}{2})^+} 3x^2 - x - 1 = \frac{1}{4}$$

et

$$\lim_{x \rightarrow \left(\frac{1}{2}\right)^+} 4x^2 - 1 = 0^+$$

Donc :

$$\lim_{x \rightarrow \left(\frac{1}{2}\right)^+} \frac{3x^2 - x - 1}{4x^2 - 1} = +\infty$$

Car :

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} (2x - 5\sqrt{x})$$

On a :

$$\lim_{x \rightarrow +\infty} (2x - 5\sqrt{x}) = \lim_{x \rightarrow +\infty} \sqrt{x}(2\sqrt{x} - 5)$$

Donc :

$$\lim_{x \rightarrow +\infty} (2x - 5\sqrt{x}) = +\infty$$

Car :

$$\lim_{x \rightarrow +\infty} \sqrt{x} = +\infty$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} \left( \sqrt{9x^2 + 3x - 5} + x + 3 \right)$$

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} \left( \sqrt{9x^2 + 3x - 5} + x + 3 \right) &= \lim_{x \rightarrow -\infty} \left( \sqrt{x^2 \left( 9 + \frac{3}{x} - \frac{5}{x^2} \right)} + x + 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left( |x| \sqrt{\left( 9 + \frac{3}{x} - \frac{5}{x^2} \right)} + x + 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left( -x \sqrt{\left( 9 + \frac{3}{x} - \frac{5}{x^2} \right)} + x + 3 \right) \\ &= \lim_{x \rightarrow -\infty} \left[ x \left( -\sqrt{\left( 9 + \frac{3}{x} - \frac{5}{x^2} \right)} + 1 \right) + 3 \right] \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -\infty} \left( \sqrt{9x^2 + 3x - 5} + x + 3 \right) = +\infty$$

Car :

$$\lim_{x \rightarrow -\infty} \frac{3}{x} = \lim_{x \rightarrow -\infty} \frac{5}{x} = 0 \quad \text{et} \quad \lim_{x \rightarrow -\infty} x = -\infty$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} (\sqrt{x+3} - x)$$

On a :

$$\begin{aligned}\lim_{x \rightarrow +\infty} (\sqrt{x+3} - x) &= \lim_{x \rightarrow +\infty} \sqrt{x \left(1 + \frac{3}{x}\right)} - x \\ &= \lim_{x \rightarrow +\infty} \sqrt{x} \sqrt{\left(1 + \frac{3}{x}\right)} - \sqrt{x^2} \\ &= \lim_{x \rightarrow +\infty} \sqrt{x} \left( \sqrt{\left(1 + \frac{3}{x}\right)} - \sqrt{x} \right)\end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} (\sqrt{x+3} - x) = -\infty$$

Car :

$$\lim_{x \rightarrow +\infty} \sqrt{x} = +\infty \text{ et } \lim_{x \rightarrow +\infty} \frac{3}{x} = 0$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4)$$

On a :

$$\begin{aligned}\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4) &= \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - (x - 4)) \\ &= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^2 + 5x + 2} - (x - 4))(\sqrt{x^2 + 5x + 2} + (x - 4))}{(\sqrt{x^2 + 5x + 2} + (x - 4))} \\ &= \lim_{x \rightarrow +\infty} \frac{13 - \frac{14}{x}}{\sqrt{1 + \frac{5}{x} + \frac{2}{x^2}} + 1 - \frac{4}{x}}\end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 5x + 2} - x + 4) = \frac{13}{2}$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x})$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x}) &= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})}{(\sqrt{x+1} + \sqrt{x})} \\ &= \lim_{x \rightarrow +\infty} \frac{1}{(\sqrt{x+1} + \sqrt{x})} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x}) = 0$$

Car :

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} + \sqrt{x}) = +\infty$$

et

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3)$$

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3) &= \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + (2x - 3)) \\ &= \lim_{x \rightarrow -\infty} \frac{(\sqrt{4x^2 + x + 3} + (2x - 3))(\sqrt{4x^2 + x + 3} - (2x - 3))}{(\sqrt{4x^2 + x + 3} - (2x - 3))} \\ &= \lim_{x \rightarrow -\infty} \frac{x \left(13 - \frac{6}{x}\right)}{x \left(-\sqrt{4 + \frac{1}{x} + \frac{3}{x^2}} - 2 + \frac{3}{x}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{13 - \frac{6}{x}}{-\sqrt{4 + \frac{1}{x} + \frac{3}{x^2}} - 2 + \frac{3}{x}} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + x + 3} + 2x - 3) = -\frac{13}{4}$$

➤ Calculons :

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x} &= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 \left(1 + \frac{5}{x} - \frac{3}{x^2}\right)}}{x} \\ &= \lim_{x \rightarrow -\infty} \frac{-x \times \sqrt{\left(1 + \frac{5}{x} - \frac{3}{x^2}\right)}}{x} \\ &= \lim_{x \rightarrow -\infty} -\sqrt{\left(1 + \frac{5}{x} - \frac{3}{x^2}\right)} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 5x - 3}}{x} = -1$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{4x^2 + 3}{x^2 + 8x}}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{4x^2 + 3}{x^2 + 8x} &= \lim_{x \rightarrow +\infty} \frac{4x^2}{x^2} \\ &= \lim_{x \rightarrow +\infty} 4 \\ &= 4 \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{4x^2 + 3}{x^2 + 8x}} = 2$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2}$$

On a :

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \times \sqrt{1 + \frac{1}{x}}}{x \left(1 + \frac{2}{x}\right)}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2} = \lim_{x \rightarrow +\infty} \frac{\sqrt{1 + \frac{1}{x}}}{\sqrt{x} \left(1 + \frac{2}{x}\right)}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x+1}}{x+2} = 0$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{x + 6\sqrt{x}}{2x + \sqrt{x} - 3}$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{x + 6\sqrt{x}}{2x + \sqrt{x} - 3} &= \lim_{x \rightarrow +\infty} \frac{x \left(1 + \frac{6}{\sqrt{x}}\right)}{x \left(2 + \frac{1}{\sqrt{x}} - \frac{3}{x}\right)} \\ &= \lim_{x \rightarrow +\infty} \frac{1 + \frac{6}{\sqrt{x}}}{2 + \frac{1}{\sqrt{x}} - \frac{3}{x}} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{x + 6\sqrt{x}}{2x + \sqrt{x} - 3} = \frac{1}{2}$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{1}{x} \left( \sqrt{3x^2 + 1} - \sqrt{x^2 + x} \right)$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{1}{x} \left( \sqrt{3x^2 + 1} - \sqrt{x^2 + x} \right) &= \lim_{x \rightarrow +\infty} \frac{1}{x} \left( \sqrt{x^2 \left(3 + \frac{1}{x^2}\right)} - \sqrt{x^2 \left(1 + \frac{1}{x}\right)} \right) \\ &= \lim_{x \rightarrow +\infty} \left( \sqrt{\left(3 + \frac{1}{x^2}\right)} - \sqrt{\left(1 + \frac{1}{x}\right)} \right) \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{1}{x} \left( \sqrt{3x^2 + 1} - \sqrt{x^2 + x} \right) = \sqrt{3} - 1$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \left( \sqrt{x^4 - x^3} - x^2 \right)$$

On a :

$$\lim_{x \rightarrow +\infty} \left( \sqrt{x^4 - x^3} - x^2 \right) = \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^4 - x^3} - x^2)(\sqrt{x^4 - x^3} + x^2)}{(\sqrt{x^4 - x^3} + x^2)}$$

Donc :

$$\begin{aligned}\lim_{x \rightarrow +\infty} \left( \sqrt{x^4 - x^3} - x^2 \right) &= \lim_{x \rightarrow +\infty} \frac{-x^3}{x^2 \left( \sqrt{1 - \frac{1}{x}} + 1 \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{-x}{\sqrt{1 - \frac{1}{x}} + 1}\end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \left( \sqrt{x^4 - x^3} - x^2 \right) = -\infty$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2 + x} - 3x^2 + x}{5\sqrt{x^2 + 3}}$$

On a :

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2 + x} - 3x^2 + x}{5\sqrt{x^2 + 3}} &= \lim_{x \rightarrow +\infty} \left( \frac{3x\sqrt{x^2 + x} - 3x^2}{5\sqrt{x^2 + 3}} + \frac{x}{5\sqrt{x^2 + 3}} \right) \\ &= \lim_{x \rightarrow +\infty} \left( \frac{(3x\sqrt{x^2 + x} - 3x^2)(3x\sqrt{x^2 + x} + 3x^2)}{5\sqrt{x^2 + 3}(3x\sqrt{x^2 + x} + 3x^2)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right)\end{aligned}$$

➤ Calculons :

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2 + x} - 3x^2 + x}{5\sqrt{x^2 + 3}} &= \lim_{x \rightarrow +\infty} \left( \frac{(3x\sqrt{x^2 + x} - 3x^2)(3x\sqrt{x^2 + x} + 3x^2)}{5\sqrt{x^2 + 3}(3x\sqrt{x^2 + x} + 3x^2)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right) \\&= \lim_{x \rightarrow +\infty} \left( \frac{9x^3}{5x\sqrt{1 + \frac{3}{x^2}} \times x^2 \left( 3\sqrt{1 + \frac{1}{x}} + 3 \right)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right) \\&= \lim_{x \rightarrow +\infty} \left( \frac{9}{5 \times \sqrt{1 + \frac{3}{x^2}} \left( 3\sqrt{1 + \frac{1}{x}} + 3 \right)} + \frac{1}{5 \times \sqrt{1 + \frac{3}{x^2}}} \right)\end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \frac{3x\sqrt{x^2 + x} - 3x^2 + x}{5\sqrt{x^2 + 3}} = \frac{1}{2}$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \left( \sqrt{\frac{x^3}{x+1}} - x \right)$$

On a :

$$\lim_{x \rightarrow +\infty} \left( \sqrt{\frac{x^3}{x+1}} - x \right) = \lim_{x \rightarrow +\infty} \frac{\left( \sqrt{\frac{x^3}{x+1}} - x \right) \left( \sqrt{\frac{x^3}{x+1}} + x \right)}{\left( \sqrt{\frac{x^3}{x+1}} + x \right)}$$

$$= \lim_{x \rightarrow +\infty} \frac{\frac{x^3}{x+1} - x^2}{\left( \sqrt{\frac{x^3}{x(1 + \frac{1}{x})}} + x \right)}$$

$$= \lim_{x \rightarrow +\infty} \frac{\frac{-x^2}{x+1}}{\left( \sqrt{\frac{x^2}{(1 + \frac{1}{x})}} + x \right)}$$

Donc :

$$\begin{aligned}\lim_{x \rightarrow +\infty} \left( \sqrt{\frac{x^3}{x+1}} - x \right) &= \lim_{x \rightarrow +\infty} \frac{-x^2}{(x+1) \left( x \times \sqrt{\frac{1}{(1+\frac{1}{x})}} + x \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{-1}{\left(1 + \frac{1}{x}\right) \left( \sqrt{\frac{1}{(1+\frac{1}{x})}} + 1 \right)}\end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \left( \sqrt{\frac{x^3}{x+1}} - x \right) = -\frac{1}{2}$$

➤ Calculons :

$$\lim_{x \rightarrow +\infty} \left( \frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right)$$

On a :

$$\begin{aligned} \lim_{x \rightarrow +\infty} \left( \frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right) &= \lim_{x \rightarrow +\infty} \frac{x\sqrt{x+2} - x\sqrt{x-2}}{\sqrt{x-2}\sqrt{x+2}} \\ &= \lim_{x \rightarrow +\infty} \frac{(x\sqrt{x+2} - x\sqrt{x-2})(x\sqrt{x+2} + x\sqrt{x-2})}{\sqrt{x-2}\sqrt{x+2}(x\sqrt{x+2} + x\sqrt{x-2})} \\ &= \lim_{x \rightarrow +\infty} \frac{4x^2}{x^2 \times \sqrt{1 - \frac{2}{x}} \sqrt{1 + \frac{2}{x}} (\sqrt{x+2} + \sqrt{x-2})} \\ &= \lim_{x \rightarrow +\infty} \frac{4}{\sqrt{1 - \frac{2}{x}} \sqrt{1 + \frac{2}{x}} (\sqrt{x+2} + \sqrt{x-2})} \end{aligned}$$

Donc :

$$\lim_{x \rightarrow +\infty} \left( \frac{x}{\sqrt{x-2}} - \frac{x}{\sqrt{x+2}} \right) = 0$$